Department of Computer Science and Engineering

University of Barishal

Course Title: Simulation and Modeling

Course Code: CSE-4107

4th year 1st Semester Final Examination

Admission Session: 2016-2017

Time: 3 Hours **Marks:** 60

Answer any five Questions from the followings.

1. a) Define System. Write the components of a system with an example. [4]

   b) Explain the characteristics of a System. [4]

   c) Define the following terms with example: i) Open System ii) Closed System [4]

2. At a grocery store with one counter, customers arrive at random from 1 to 8 minutes apart (each of inter-arrival time has the same probability of occurrence). The service times vary from 1 to 6 minutes with the probabilities as 0.10, 0.20, 0.30, 0.25, 0.10 and 0.05 respectively. Analyze the system by simulating the arrival and service of 15 customers. [Justifying your situation and requirements, you can choose your required random values] [12]

3. a) Write short notes on the followings: [8]

   i) Weibull Distribution ii) Gamma Distribution

   iii) Geometric Distribution iv) Bernoulli Distribution

   b) A Hurricane is to hit in the country, and expected to follow poisson distribution with a mean of 0.8 per year. Find the possibility of occurring more than two hurricanes in a year. Also find the possibility of exactly one hurricane in a year. [4]

4. a) Explain over aspects and characteristics of a Queueing System. [8]

   b) Customers at a restaurant arrive in groups (one to eight persons). The number of persons (per group) for 300 customers and the relative frequencies appear are shown below. Draw empirical CDF. [4]

Arrivals per party	1	2	3	4	5	6	7	8
Frequency	30	110	45	71	12	13	7	12

5. a) Explain different methods of random number generation. [9]

   b) What are the challenges in generating pseudo random numbers. [3]

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